

E-COMMERCE SALES ANALYSIS

Data Preprocessing & Exploratory Data Analysis

Internship Project Report

Submitted by: Tabassum Gulnar
Organization: CodeOrbit Tech
Role: Machine Learning Intern
Project: E-Commerce Sales Analysis

1. Project Overview

The objective of this project is to analyse an e-commerce sales dataset using Python and Exploratory Data Analysis (EDA). The project focuses on understanding sales trends, identifying profitable categories, comparing regional performance and examining data quality.

2. Project Objectives

- Understand and explore an e-commerce dataset.
- Check missing values and duplicate records.
- Perform data preprocessing and date conversion.
- Analyse sales and profit across categories and regions.
- Identify monthly sales trends.
- Visualize findings using Python libraries.

3. Tools and Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab and the Kaggle Superstore dataset.

4. Data Preprocessing and Methodology

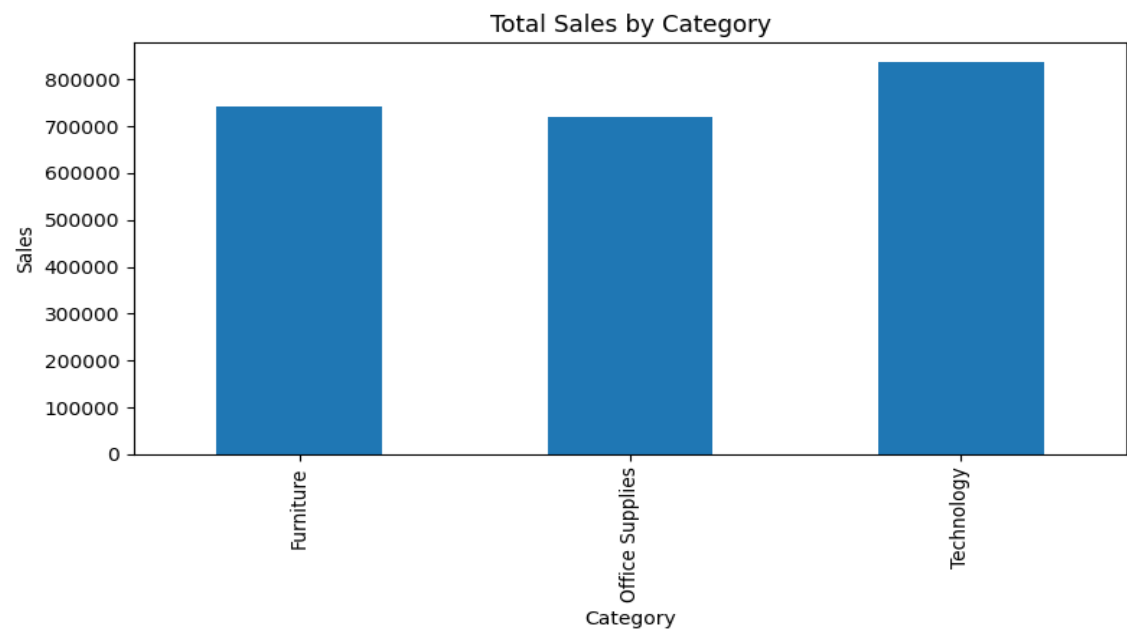
The dataset was loaded using Pandas. Its structure, data types, missing values and duplicate records were examined. Order Date and Ship Date were converted into datetime format. Year and Month columns were created to support time-based analysis. The IQR method was used to identify unusual sales observations.

5. Key Results

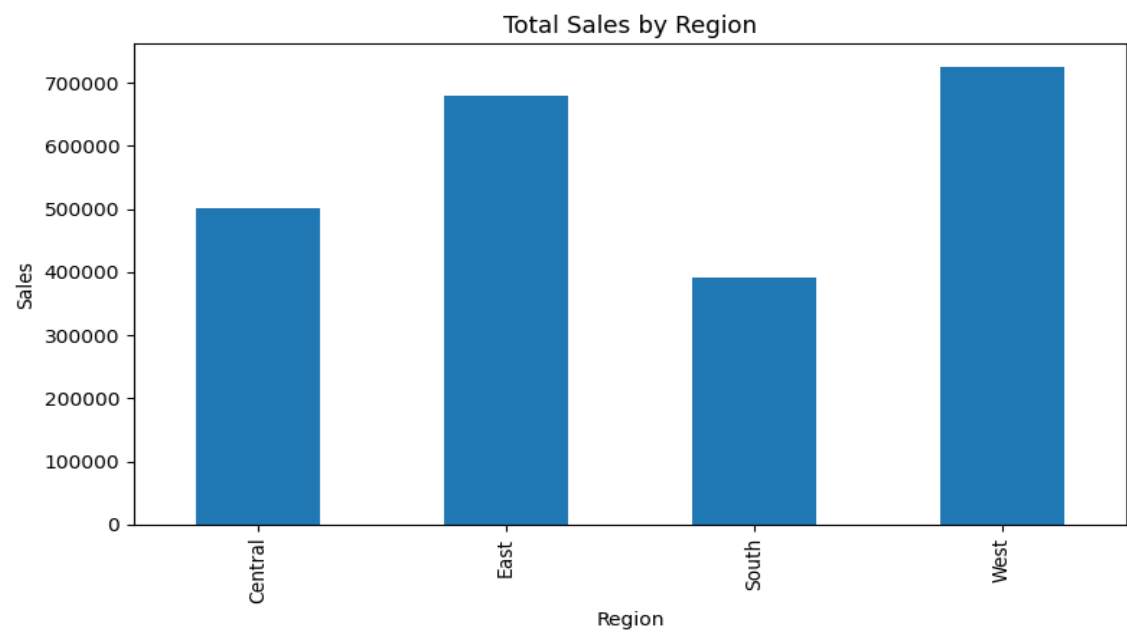
Metric	Result
Total records	9,994
Total columns after preprocessing	23
Missing values	0
Duplicate records	0
Total sales	\$2,297,200.86
Total profit	\$286,397.02
Overall profit margin	12.47%
Best-performing category	Technology
Highest sales region	West
Highest sales month	November
Sales outliers identified	1,167

6. Data Visualizations

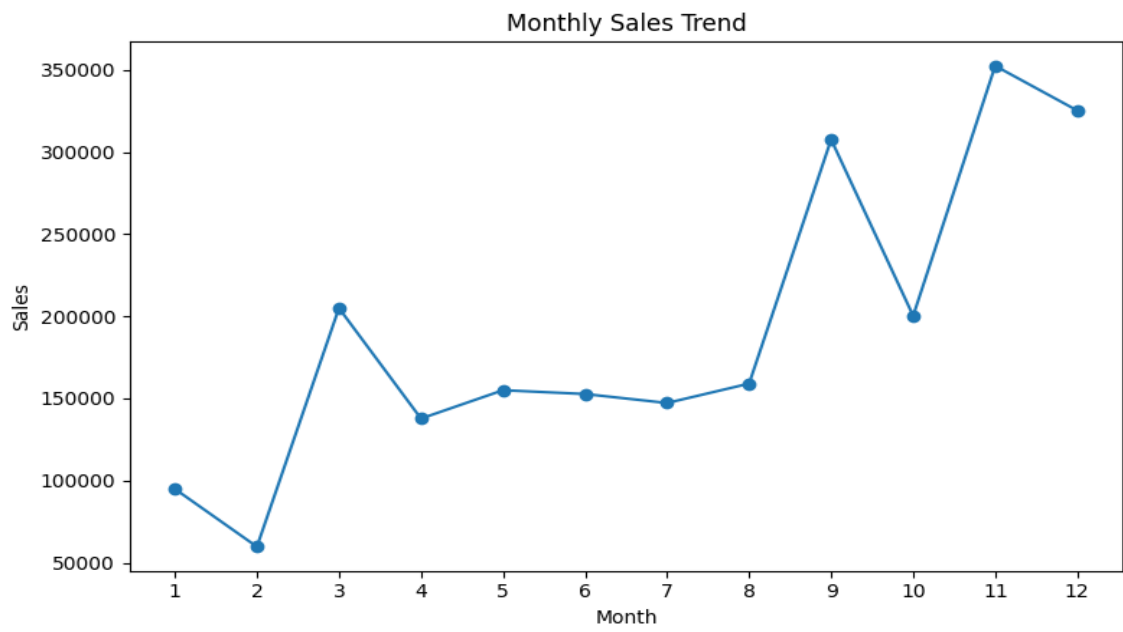
Sales by Category



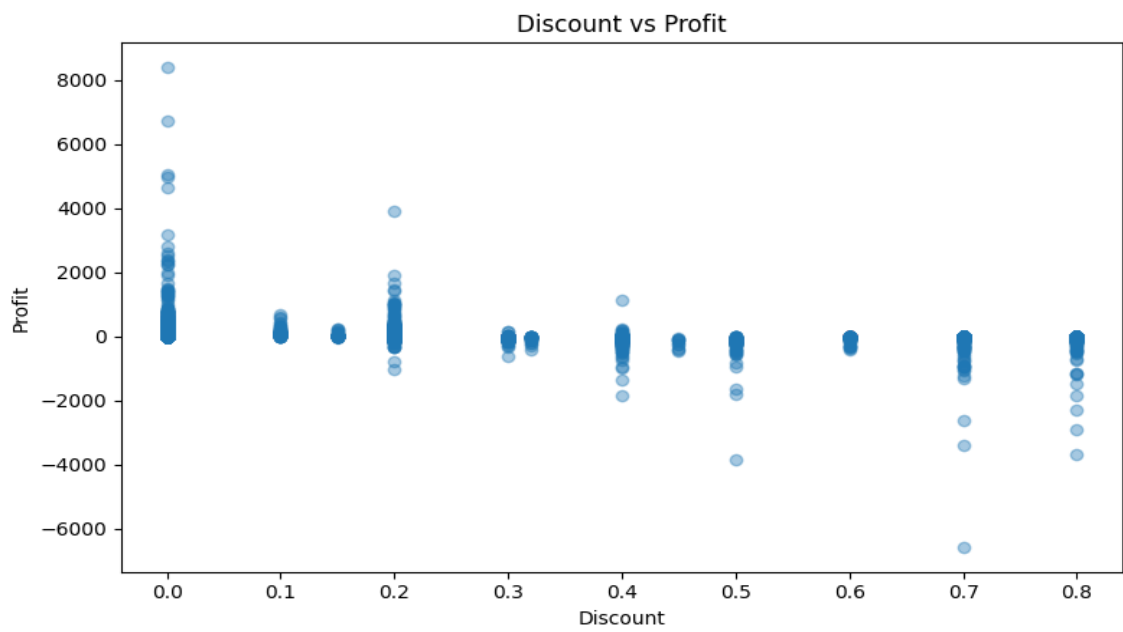
Sales by Region



Monthly Sales Trend



Discount vs Profit



7. Key Findings

- The dataset contains 9,994 records and 23 columns after preprocessing.
- No missing values or duplicate records were found.
- Technology performed best in both sales and profit.
- The West region recorded the highest total sales.
- November had the highest aggregated monthly sales across all years.
- The overall profit margin was 12.47%.
- The IQR method flagged 1,167 unusual sales observations. These observations were identified for investigation, not automatically treated as errors.

8. Conclusion

This project provided practical experience in data preprocessing, exploratory data analysis, statistical calculations and data visualization using Python. The findings highlight the contribution of the Technology category and the strong sales performance of the West region. The analysis also provides a foundation for future business intelligence and machine learning projects.

9. Future Scope

The project can be extended by developing an interactive dashboard, studying customer purchasing behaviour, investigating discount strategies and building machine learning models for sales prediction.

10. Dataset Reference

Kaggle – Superstore Dataset
<https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>